

What can I do after finding the $B^\pm$ candidate mass peak?

Continuing my first LHCb ROOT-file exercise through candidate selection and improved peak fitting

In the first part of this project, I attempted to reproduce part of the CERN and REANA $B^\pm \rightarrow J/\psi K^\pm$ example.

I opened an LHCb ROOT ntuple, extracted the `Bp1us_MM` branch, plotted the reconstructed candidate-mass distribution and fitted the visible peak using a simple Gaussian-plus-constant model.

After completing that exercise, I wanted to continue working with the same dataset rather than immediately move to a new project. However, I did not know what a sensible next analysis would be, what the additional branches represented or what kinds of selections and fitting models would normally be investigated.

I therefore asked ChatGPT to suggest a sequence of further data-analysis tasks. I treated these as programming and statistical exercises rather than as an attempt to reproduce a complete LHCb analysis.

The continuation eventually became two tasks: testing a vertex-quality selection and exploring progressively more flexible models for the selected candidate-mass distribution.

Where this report begins

The earlier part of this project stopped after fitting the visible $B\pi_{\text{plus}}^0$ peak with a Gaussian-plus-constant model.

That analysis gave a fitted centre of approximately

$$5279.4 \text{ MeV}/c^2$$

and a fitted Gaussian width of approximately

$$17.75 \text{ MeV}/c^2.$$

The purpose of this continuation was to go one step beyond that first exercise.

I divided the continuation into two stages:

Analysis 3: testing a reconstructed vertex-quality selection,

Analysis 4: fitting the selected mass distribution using several signal-plus-background models and comparing how well they described the histogram.

Important limitation

This remains a programming and data-analysis exercise

I am not presenting this work as an independent particle-physics analysis or as a scientifically validated measurement.

I did not design the ntuple, reconstruct the candidates or independently verify which candidate-selection quantities and fit models would be appropriate for a real LHCb measurement.

The tasks, including the vertex-quality selection and the progression of fit models, were suggested through discussions with ChatGPT after I asked how I could extend my first ROOT-file exercise.

My aim was to practise handling related arrays, applying Boolean conditions, comparing distributions, constructing multi-component models, examining residuals and calculating simple goodness-of-fit diagnostics.

I therefore describe the additional variables and models only to the level needed to explain what I did computationally.

Terms such as “signal”, “background”, “vertex quality” and “efficiency” are used cautiously throughout this report and should not be interpreted as the result of a validated LHCb analysis.

Data used

I continued using the same CERN Open Data sample as in the previous part:

$$B^\pm \rightarrow J/\psi(\rightarrow \mu^+ \mu^-) K^\pm$$

Ntuples 4530 — 2017 Magnet Up.

The dataset is available through:

[CERN Open Data record 93949](#)

The parent ntuple collection is identified by:

[10.7483/OOPENDATA.LHCB.EBYF.C7OY](#)

Table 1: Dataset and branches used in the continuation.

Item	Choice used here
Experiment	LHCb
Data-taking period	2017
Magnet configuration	Magnet Up
ROOT tree	Btree/DecayTree
Mass branch	Bplus_MM
Vertex-fit branch	Bplus_ENDVERTEX_CHI2
Degrees-of-freedom branch	Bplus_ENDVERTEX_NDOF
Derived quantity	Bplus_ENDVERTEX_CHI2/Bplus_ENDVERTEX_NDOF

Analysis 3: Testing a vertex-quality selection

Why I tried this

The ROOT tree contains more than the reconstructed candidate mass. Each candidate also has quantities related to how it was reconstructed.

ChatGPT suggested that I investigate the branches

`Bplus_ENDVERTEX_CHI2` and `Bplus_ENDVERTEX_NDOF`.

I understood these operationally as quantities associated with a fitted common vertex for the tracks used to form a $B^\pm$ candidate.

The ratio

$$\frac{\chi^2}{\text{NDOF}} = \frac{\text{Bplus_ENDVERTEX_CHI2}}{\text{Bplus_ENDVERTEX_NDOF}}$$

was treated as a candidate-quality quantity. Lower values represent better agreement with the fitted common vertex within the reconstruction.

I did not derive this choice from my own knowledge of LHCb analysis. I used it because it was suggested as a manageable next exercise involving multiple related branches and a candidate-selection condition.

The computational question

If I retain only candidates with sufficiently low χ^2/NDOF , does the reconstructed mass distribution change uniformly, or are candidates near the visible peak retained more often than candidates away from it?

Keeping the arrays aligned

Each entry in the tree represents one reconstructed candidate. The mass, chi-squared value and number of degrees of freedom belonging to one candidate therefore occupy the same index in their respective arrays.

This meant that I could not remove values independently from each array. A shared Boolean mask had to be applied so that

$$\text{mass}[i], \quad \text{chi2}[i], \quad \text{ndof}[i]$$

continued to refer to the same candidate.

The validity conditions were:

- finite Bplus_MM,
- finite Bplus_ENDVERTEX_CHI2,
- finite Bplus_ENDVERTEX_NDOF,
- non-negative chi-squared,
- and $\text{NDOF} > 0$.

The number of entries before and after this validity check was

1 335 748.

All three arrays contained the same number of valid entries. Therefore, the validity mask removed no candidate from this particular sample.

Table 2: Array lengths before applying a vertex-quality threshold.

Array	Number of entries
Finite raw Bplus_MM values	1 335 748
Mass values with valid vertex quantities	1 335 748
Valid χ^2/NDOF values	1 335 748

Thresholds tested

I tested three upper limits:

$$\frac{\chi^2}{\text{NDOF}} < 5, \quad \frac{\chi^2}{\text{NDOF}} < 3, \quad \frac{\chi^2}{\text{NDOF}} < 2.$$

These were not treated as established or optimised LHCb selections. They were used as three levels of strictness so that I could observe the trade-off between retaining candidates and rejecting candidates with larger vertex-fit ratios.

Table 3: Decision log for Analysis 3.

Part	Choice	Value or method	Reason
Selection quantity	Vertex-fit ratio	χ^2/NDOF	Suggested by ChatGPT as a candidate-quality quantity that could be tested using the available branches.
Loose threshold	Maximum accepted ratio	< 5	Used as the least restrictive of the three tested conditions.
Intermediate threshold	Maximum accepted ratio	< 3	Used to observe how a stricter condition changed candidate retention.
Strict threshold	Maximum accepted ratio	< 2	Used to test the effect of a substantially more restrictive condition.
Mass comparison	Displayed interval	Approximately 5220–5360 MeV/ c^2	Covered the visible peak and regions on both sides of it.
Histogram comparison	Bin edges	Identical for all samples	Ensured that before-and-after bin counts could be compared directly.
Overall retention	Calculation	$N_{\text{after}}/N_{\text{before}}$	Measured the fraction of valid candidates retained by each condition.
Bin-wise retention	Calculation	$N_{\text{after},i}/N_{\text{before},i}$	Showed whether the selection acted uniformly across reconstructed mass.

Overall candidate retention

The percentage retained by each threshold was calculated using

$$\text{retention} = \frac{N_{\text{after cut}}}{N_{\text{before cut}}} \times 100.$$

The results were:

Table 4: Overall candidate retention for the three tested thresholds.

Vertex-quality condition	Candidates retained	Candidates rejected
$\chi^2/\text{NDOF} < 5$	87.33%	12.67%
$\chi^2/\text{NDOF} < 3$	80.07%	19.93%
$\chi^2/\text{NDOF} < 2$	69.91%	30.09%

As expected, lowering the threshold made the selection stricter.

However, these percentages alone did not show whether the removed candidates came mainly from the visible peak, mainly from the regions away from the peak, or from both at similar rates.

Comparing the mass distributions

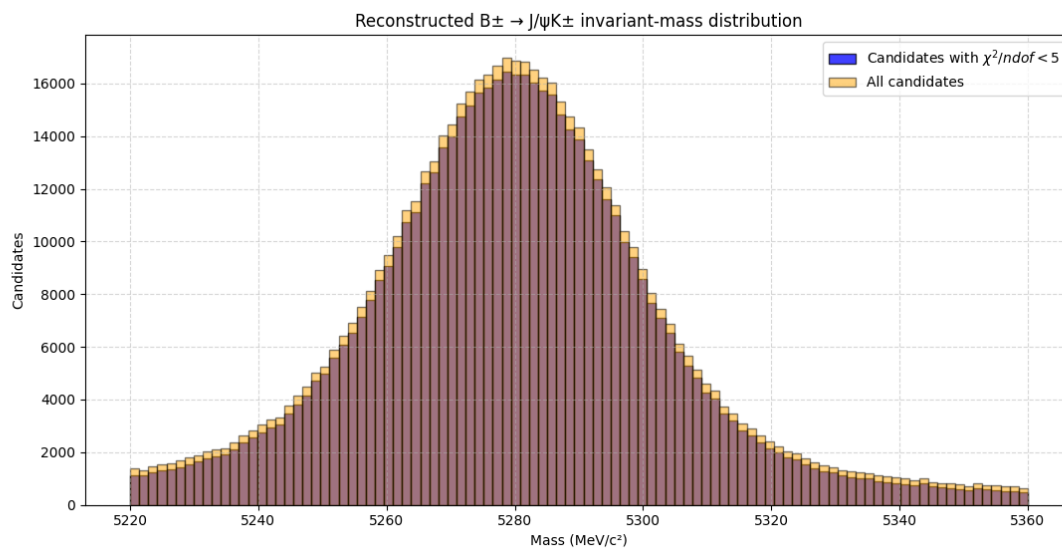


Figure 1: Reconstructed $B^\pm$ candidate-mass distribution before and after applying $\chi^2/\text{NDOF} < 5$. The selected distribution retains 87.33% of all valid candidates.

The two histograms initially appeared similar because the location and general shape of the visible peak remained almost unchanged.

The difference became clearer when stricter thresholds were applied.

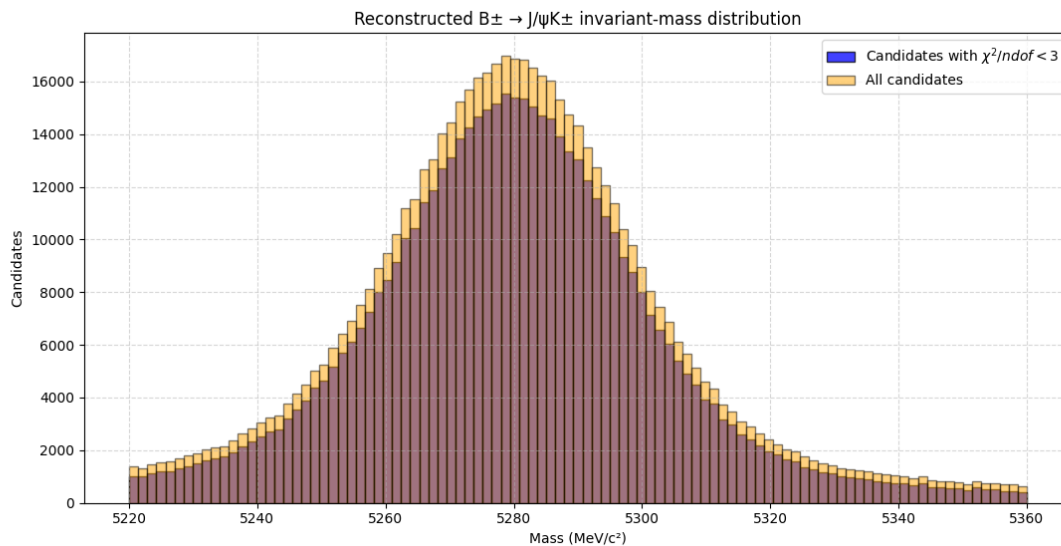


Figure 2: Reconstructed candidate-mass distribution before and after applying $\chi^2/\text{NDOF} < 3$. The selected sample retains 80.07% of the candidates.

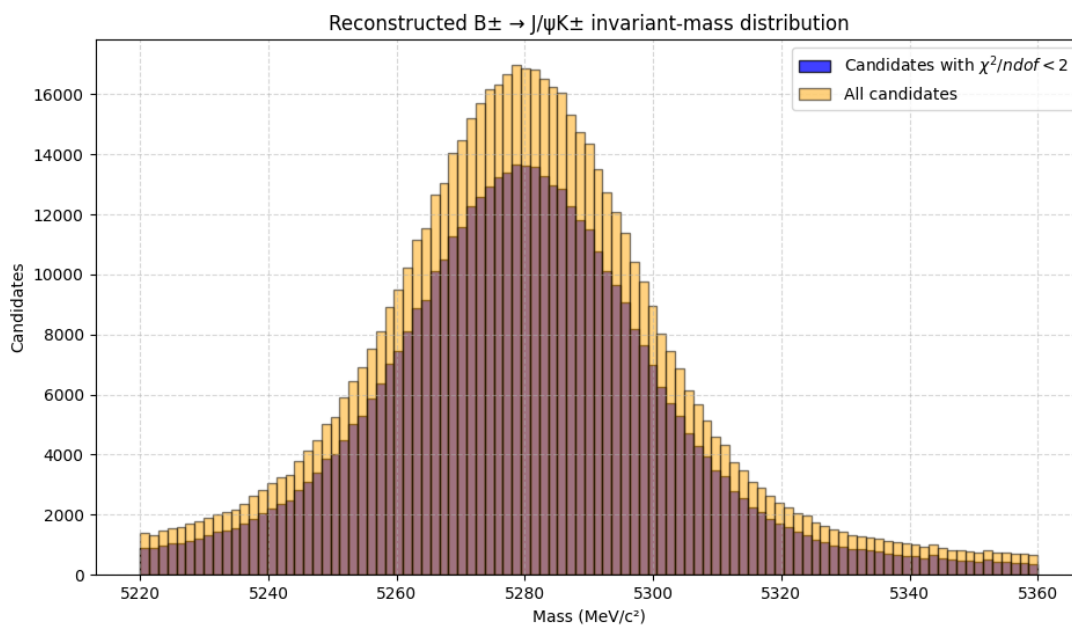


Figure 3: Reconstructed candidate-mass distribution before and after applying $\chi^2/\text{NDOF} < 2$. The selected sample retains 69.91% of the candidates.

The stricter selections reduced the histogram more strongly, but an overlaid histogram did not provide a direct numerical view of where the largest relative changes occurred.

I therefore calculated the retention separately inside every mass bin.

Retention as a function of reconstructed mass

For each mass bin i , I calculated

$$R_i = \frac{N_{\text{after},i}}{N_{\text{before},i}},$$

where $N_{\text{before},i}$ is the original number of candidates in the bin and $N_{\text{after},i}$ is the number remaining after the chosen vertex-quality condition.

A value of

$$R_i = 1$$

would mean that every candidate in that bin survived. A lower value means that a larger fraction was removed.

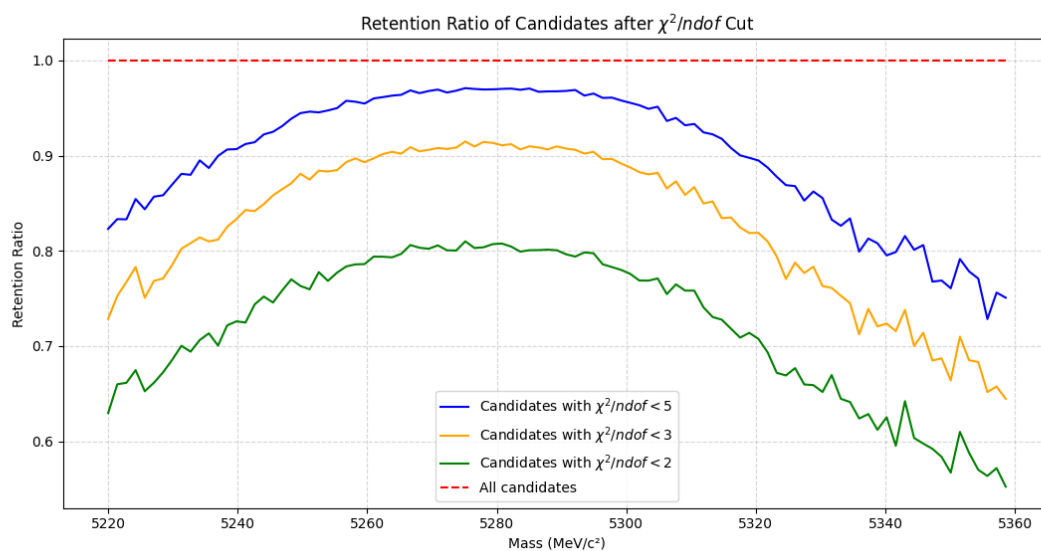


Figure 4: Candidate retention as a function of reconstructed mass for the three tested vertex-quality thresholds. The dashed horizontal line at one represents the original uncut candidate count in each bin.

The retention was not constant across reconstructed mass.

For the $\chi^2/\text{NDOF} < 5$ condition, retention reached approximately 96–97% near the visible peak around 5280 MeV/ c^2 , while it was lower toward the outer parts of the displayed interval.

The same pattern became more pronounced for the stricter thresholds.

Table 5: Approximate behaviour visible in the bin-wise retention plot.

Condition	Retention near the visible peak	Retention toward the outer mass regions
$\chi^2/\text{NDOF} < 5$	Approximately 96–97%	Approximately 75–85%, depending on the side and bin
$\chi^2/\text{NDOF} < 3$	Approximately 90–91%	Approximately 64–75%
$\chi^2/\text{NDOF} < 2$	Approximately 80%	Approximately 55–67%

These values are approximate summaries of the plotted bin ratios rather than fitted physical efficiencies.

Choosing the baseline condition

The strictest condition removed the largest number of candidates away from the peak, but it also removed a noticeably larger fraction near the peak.

The condition

$$\frac{\chi^2}{\text{NDOF}} < 5$$

gave the least aggressive compromise among the tested values:

- it retained 87.33% of all candidates,
- it retained approximately 96–97% near the visible peak,
- and it removed a larger fraction toward the outer parts of the displayed mass interval.

I therefore selected

$$\chi^2/\text{NDOF} < 5$$

as the baseline condition for Analysis 4.

Result of Analysis 3

The vertex-quality condition did not simply multiply the complete mass histogram by one constant factor.

Candidates near the visible $B^\pm$ peak were retained more frequently than candidates in the regions farther from the peak.

Among the three tested thresholds, the condition below five preserved the largest fraction near the peak while still removing more candidates outside it.

What Analysis 3 does and does not show

Supported by this exercise	Not established by this exercise
The three tested vertex-quality thresholds retain different fractions of the candidate sample.	The tested thresholds are the standard or optimal choices for an LHCb measurement.
Retention depends on reconstructed mass and is highest near the visible peak.	Every candidate near the peak is a genuine $B^\pm$ decay.
The condition below five preserves more of the peak region than the outer regions.	The plotted peak-region retention is the true signal efficiency.
The condition below two removes substantially more candidates than the condition below five.	The strictest cut necessarily gives the best final statistical result.
The selected distribution is visibly reshaped by the selection.	The selection has been fully studied for possible reconstruction or detector-dependent effects.

Why I do not call this a signal-efficiency measurement

The visible peak region still contains an unknown mixture of the peak component and background candidates.

The fraction retained in that region is therefore described as **peak-region retention**, not as a measured efficiency for genuine $B^\pm$ decays.

Analysis 4: Modelling the selected mass distribution

Analysis 3 left me with a baseline selected sample satisfying

$$\chi^2/\text{NDOF} < 5.$$

The next question was how to describe the mass distribution of this selected sample more carefully.

In the first part of the project I had used

Gaussian + constant background.

That model was deliberately simple.

For Analysis 4 I wanted to separate the visible peak-like component from the smoothly varying candidate background and test whether increasing the flexibility of the model improved the description of the data.

The question I used

How much does the fit improve when I replace the original Gaussian-plus-constant model with progressively more flexible peak-plus-background models?

Histogram used for the fits

For all model comparisons in this analysis I kept the same:

- candidate selection: $\chi^2/\text{NDOF} < 5$,
- mass interval: 5220–5360 MeV/ c^2 ,
- and histogram binning: 100 equally spaced bin edges, producing 99 fitted bins.

Keeping these choices fixed meant that changes in the fitted result could be associated with changes in the mathematical model rather than with a simultaneous change in binning or data selection.

Model 1: Gaussian plus linear background

The first extension was to replace the constant background by a straight line.

The model was

$$F(x) = A \exp \left[-\frac{(x - \mu)^2}{2\sigma^2} \right] + d + e(x - 5280).$$

The fixed reference value 5280 was used only to centre the polynomial near the mass peak. This keeps the numerical values of the fitted background parameters easier to handle.

The five fitted parameters were therefore

$$A, \quad \mu, \quad \sigma, \quad d, \quad e.$$

Table 6: Gaussian-plus-linear-background fit parameters.

Parameter	Role	Fitted value	Fit uncertainty
A	Gaussian amplitude	15102.9515	47.7158
μ	Shared peak centre in MeV/ c^2	5279.7373	0.0622
σ	Gaussian width in MeV/ c^2	17.3955	0.0752
d	Background level at 5280 MeV/ c^2	1107.2099	30.9235
e	Linear background slope	-5.3918	0.4726

The negative fitted value of e allowed the background to decrease with reconstructed mass rather than remain flat.

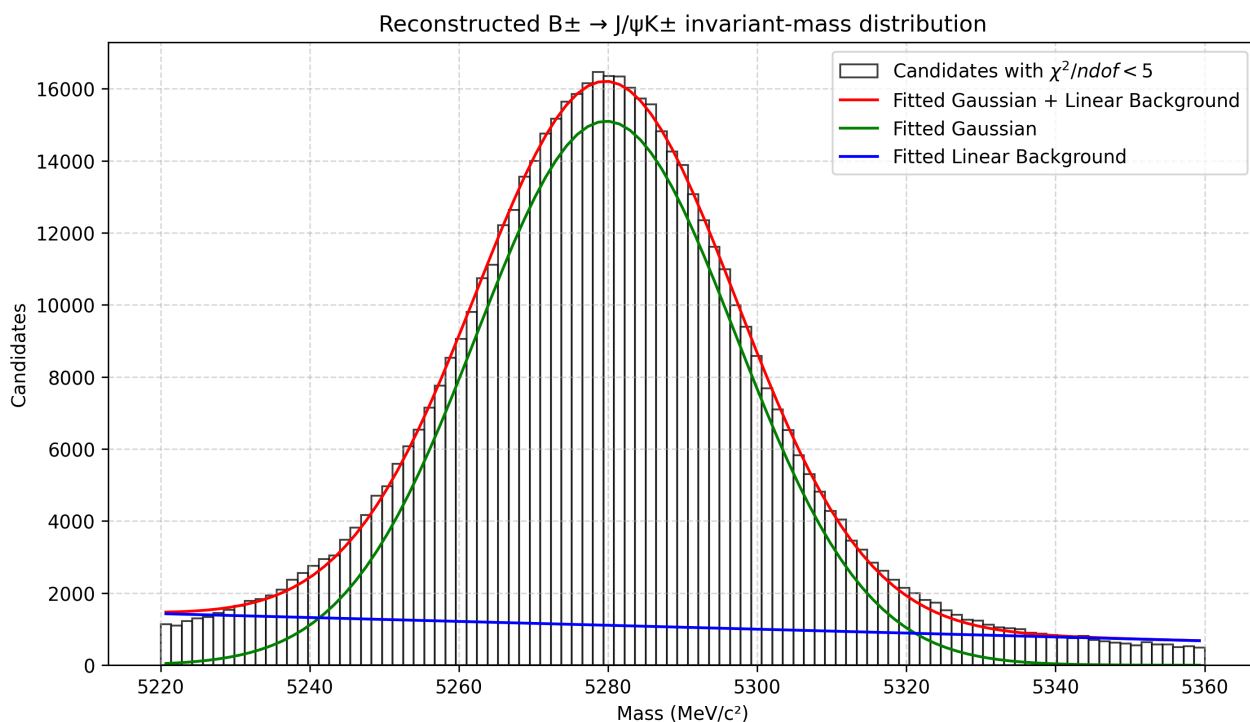


Figure 5: Selected candidate-mass histogram with the Gaussian peak component, fitted linear background and their total fitted model.

Visually, the linear background followed the falling high-mass side more closely than the constant model used in the first part of the project.

However, the residual plot showed that the remaining differences were not randomly distributed around zero.

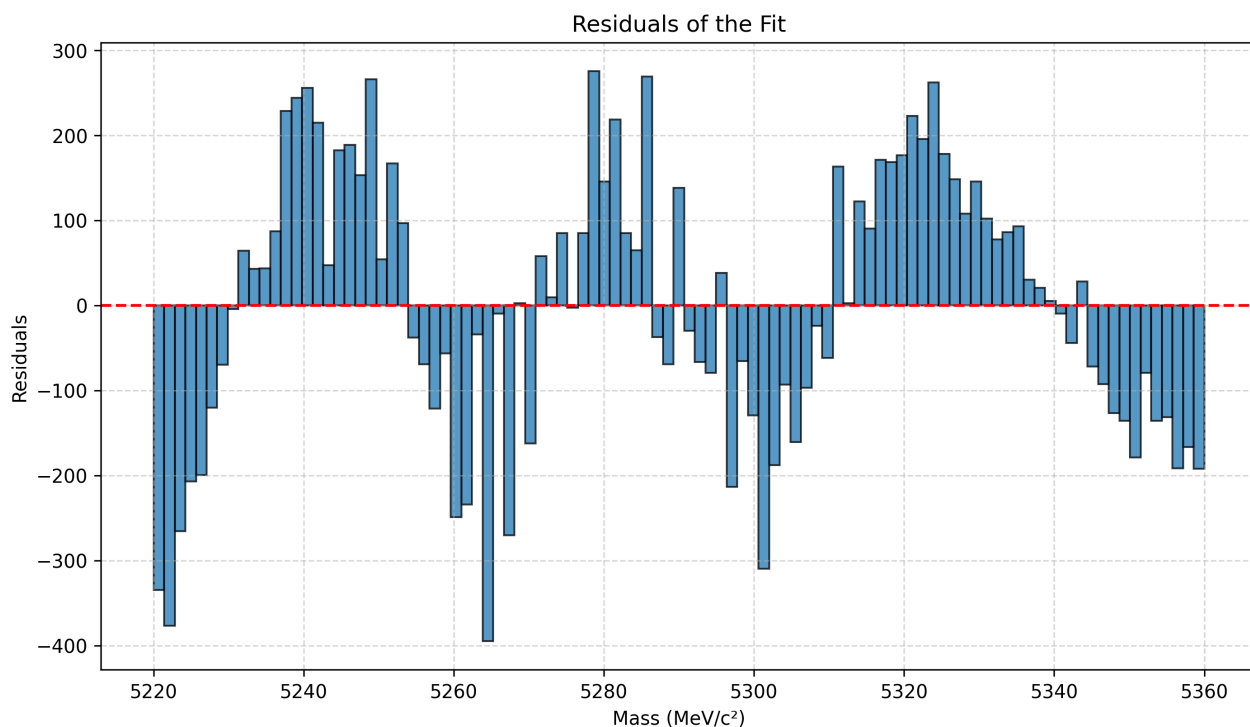


Figure 6: Raw residuals, defined as observed bin count minus fitted bin count, for the Gaussian-plus-linear-background model.

Broad regions of positive and negative residuals remained. This motivated testing a background function with curvature.

Model 2: Gaussian plus quadratic background

The next model was

$$F(x) = A \exp \left[-\frac{(x - \mu)^2}{2\sigma^2} \right] + d + e(x - 5280) + f(x - 5280)^2.$$

This introduced one additional parameter, f , controlling the quadratic curvature of the background.

The fitted parameters were:

Table 7: Gaussian-plus-quadratic-background fit parameters.

Parameter	Role	Fitted value	Fit uncertainty
A	Gaussian amplitude	14605.3515	66.5820
μ	Peak centre in MeV/c^2	5279.6380	0.0488
σ	Gaussian width in MeV/c^2	16.8574	0.0818
d	Background level at the reference mass	1673.4458	67.0133
e	Linear background term	-3.6277	0.4094
f	Quadratic background term	-0.1666	0.0190

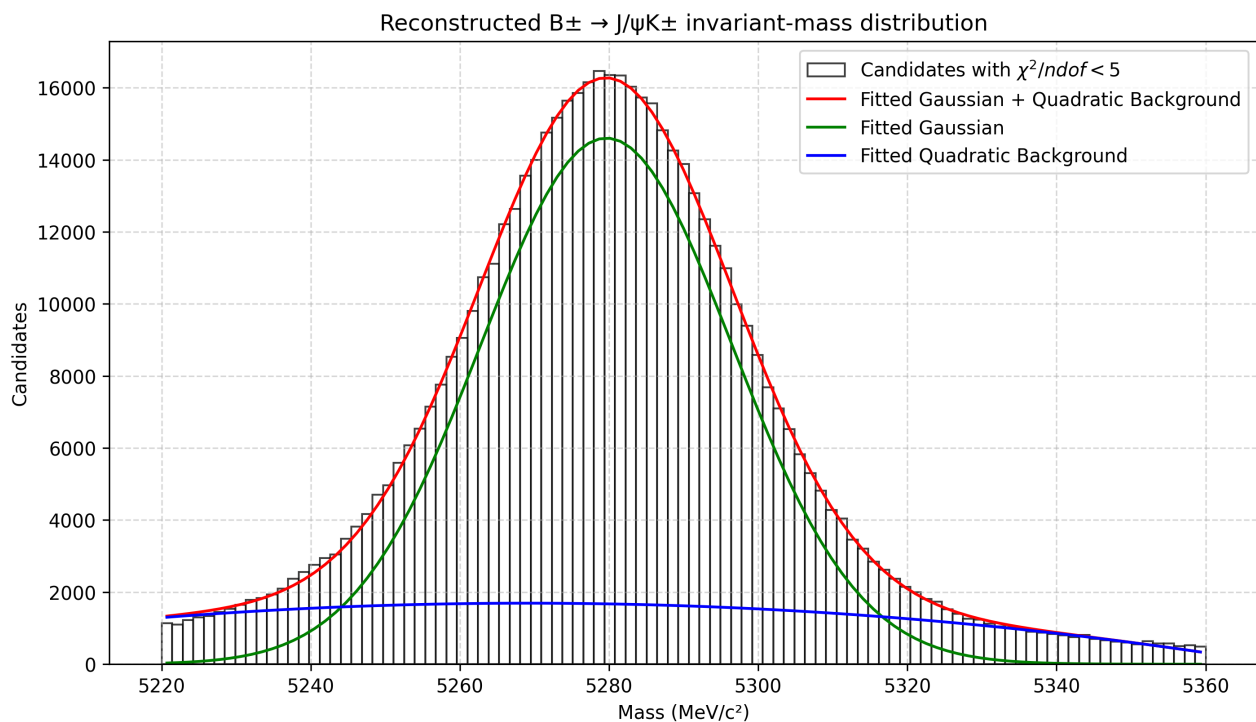


Figure 7: Selected mass distribution fitted using a Gaussian peak component and quadratic background.

The residual structure appeared less regular than in the linear model, but visible systematic deviations still remained.

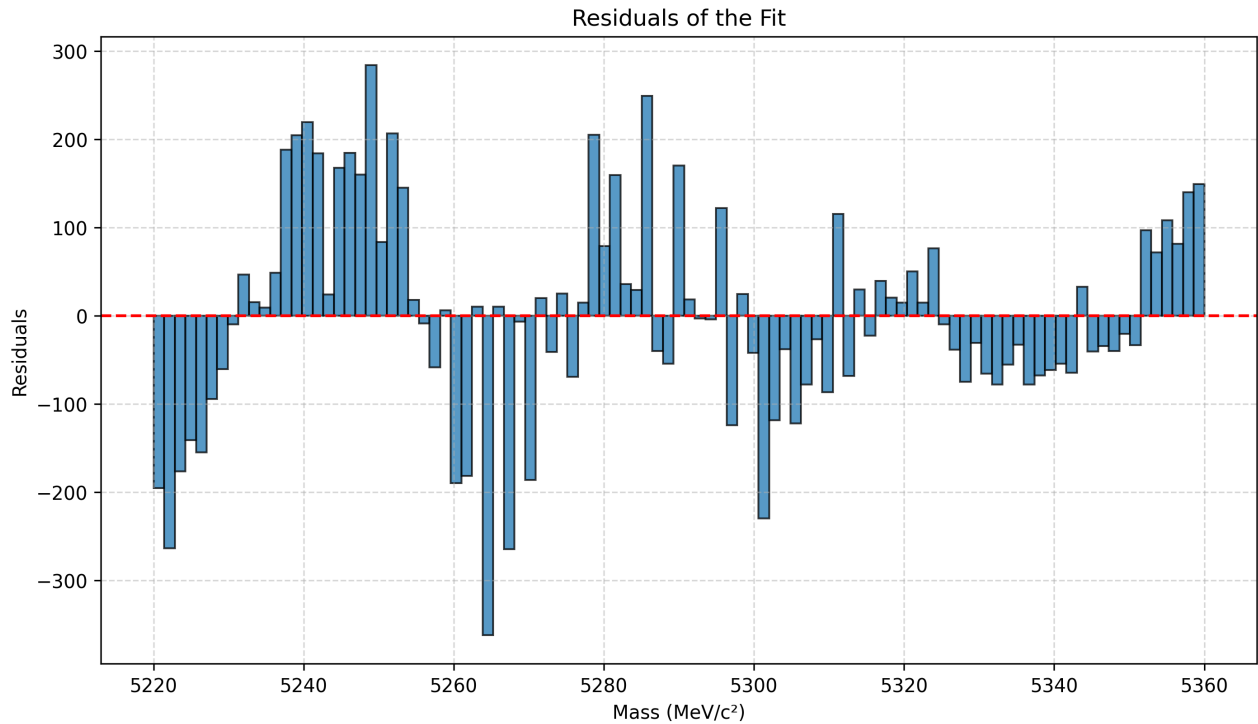


Figure 8: Raw residuals for the Gaussian-plus-quadratic-background model. The broad structure was reduced but not eliminated.

Using pulls and a goodness-of-fit diagnostic

Raw residuals measure differences in candidate counts,

$$r_i = N_i - M_i,$$

where N_i is the observed count and M_i is the fitted model prediction in bin i .

However, a difference of 100 candidates has a different relative size in a bin containing several hundred candidates than in one containing many thousands.

I therefore also calculated a Pearson-style pull,

$$p_i = \frac{N_i - M_i}{\sqrt{M_i}},$$

and the corresponding diagnostic statistic

$$\chi^2 = \sum_i \frac{(N_i - M_i)^2}{M_i} = \sum_i p_i^2.$$

For a model with k fitted parameters and 99 fitted bins, I used

$$\nu = 99 - k$$

as the number of degrees of freedom and calculated

$$\chi_{\text{red}}^2 = \frac{\chi^2}{\nu}.$$

Important limitation of this comparison

The model parameters were obtained using the default least-squares behaviour of `scipy.optimize.curve_fit`.

The Pearson statistic was calculated afterwards as a goodness-of-fit diagnostic. The fit itself was not obtained by directly minimising this Pearson chi-squared function.

I therefore use these values to compare the models in this learning exercise, not as rigorous hypothesis-test results.

For the Gaussian-plus-linear model,

$$\chi^2 = 1102.49,$$

with

$$\nu = 99 - 5 = 94,$$

giving

$$\chi_{\text{red}}^2 = 11.73.$$

For the Gaussian-plus-quadratic model,

$$\chi^2 = 568.97,$$

with

$$\nu = 99 - 6 = 93,$$

giving

$$\chi_{\text{red}}^2 = 6.12.$$

The decrease from 11.73 to 6.12 showed that allowing curvature in the background substantially improved the description of the histogram.

However, the reduced value remained far above one and the pull plot still contained structured deviations.

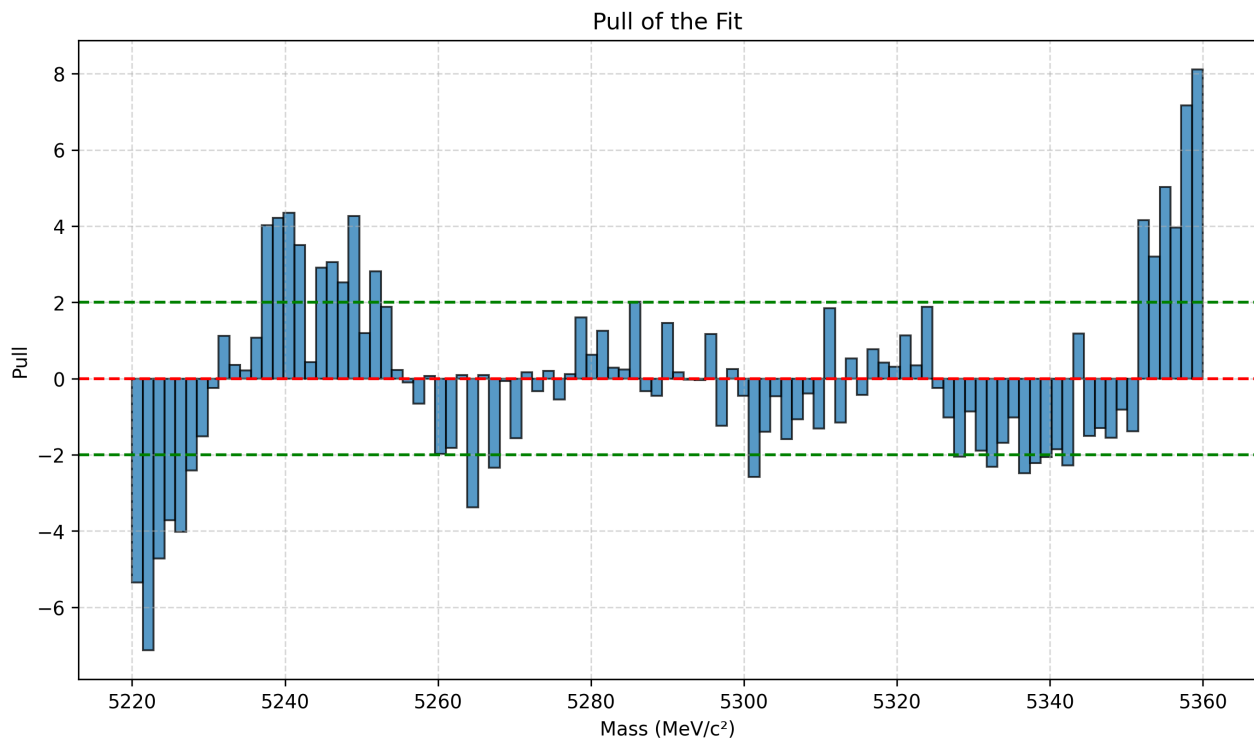


Figure 9: Pearson-style pulls for the single-Gaussian plus quadratic-background fit. The horizontal lines at $+2$ and -2 provide visual reference levels.

This suggested that changing only the background shape was not sufficient.

Model 3: Double Gaussian plus quadratic background

Instead of continuing to increase the polynomial order of the background, I next tested whether the single Gaussian itself was too restrictive.

The peak-like component was replaced by two Gaussian functions sharing the same centre:

$$G_1(x) = A_1 \exp \left[-\frac{(x - \mu)^2}{2\sigma_1^2} \right],$$

$$G_2(x) = A_2 \exp \left[-\frac{(x - \mu)^2}{2\sigma_2^2} \right].$$

The full model became

$$\begin{aligned} F(x) = & A_1 \exp \left[-\frac{(x - \mu)^2}{2\sigma_1^2} \right] \\ & + A_2 \exp \left[-\frac{(x - \mu)^2}{2\sigma_2^2} \right] \\ & + d + e(x - 5280) + f(x - 5280)^2. \end{aligned}$$

Both Gaussian components were constrained by construction to have the same fitted centre μ , while their amplitudes and widths were free to differ.

This was not interpreted as evidence for two separate physical peaks. It was simply a more flexible mathematical model for one visible peak.

Double-Gaussian fit result

The fitted parameters were:

Table 8: Double-Gaussian plus quadratic-background fit parameters.

Parameter	Role	Fitted value	Fit uncertainty
A_1	Amplitude of Gaussian 1	8987.8612	2918.2074
A_2	Amplitude of Gaussian 2	6523.6118	2654.7581
μ	Shared centre in MeV/c^2	5279.6546	0.0385
σ_1	Width of Gaussian 1 in MeV/c^2	14.6446	0.9127
σ_2	Width of Gaussian 2 in MeV/c^2	22.2717	3.0767
d	Background level at the reference mass	859.9198	314.4957
e	Linear background term	-3.7486	0.3359
f	Quadratic background term	-0.0084	0.0539

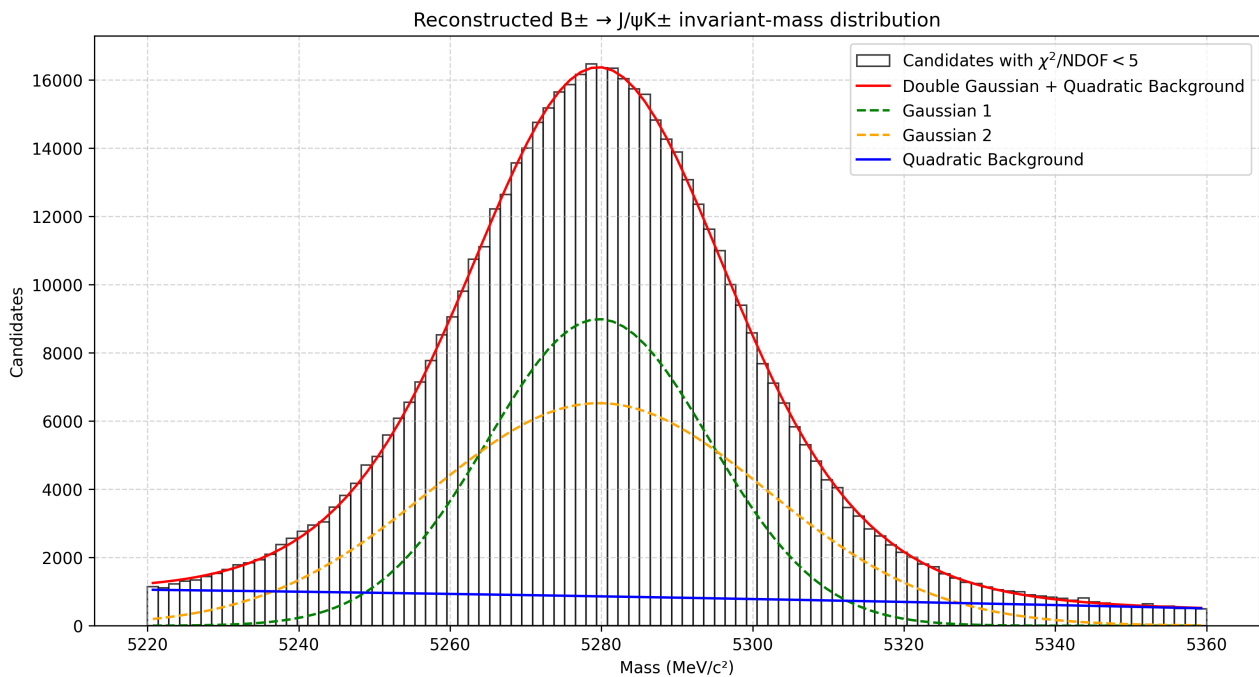


Figure 10: Selected candidate-mass distribution with two Gaussian components, the quadratic background and the complete fitted model. Both Gaussian components share the same fitted centre.

The two Gaussian widths were substantially different:

$$\sigma_1 = 14.64 \text{ MeV}/c^2$$

and

$$\sigma_2 = 22.27 \text{ MeV}/c^2.$$

The amplitudes of the two components had comparatively large covariance-derived uncertainties. I therefore do not interpret the individual Gaussian amplitudes as separately measured physical quantities.

The fitted quadratic coefficient also became

$$f = -0.0084 \pm 0.0539,$$

which is close to zero relative to its quoted fit uncertainty.

Within this particular model, much of the flexibility previously provided by the quadratic background was therefore no longer strongly required once the peak shape itself was made more flexible.

This should not be interpreted as proving that the true physical background is linear. The fitted components can be correlated with one another.

Residuals and pulls

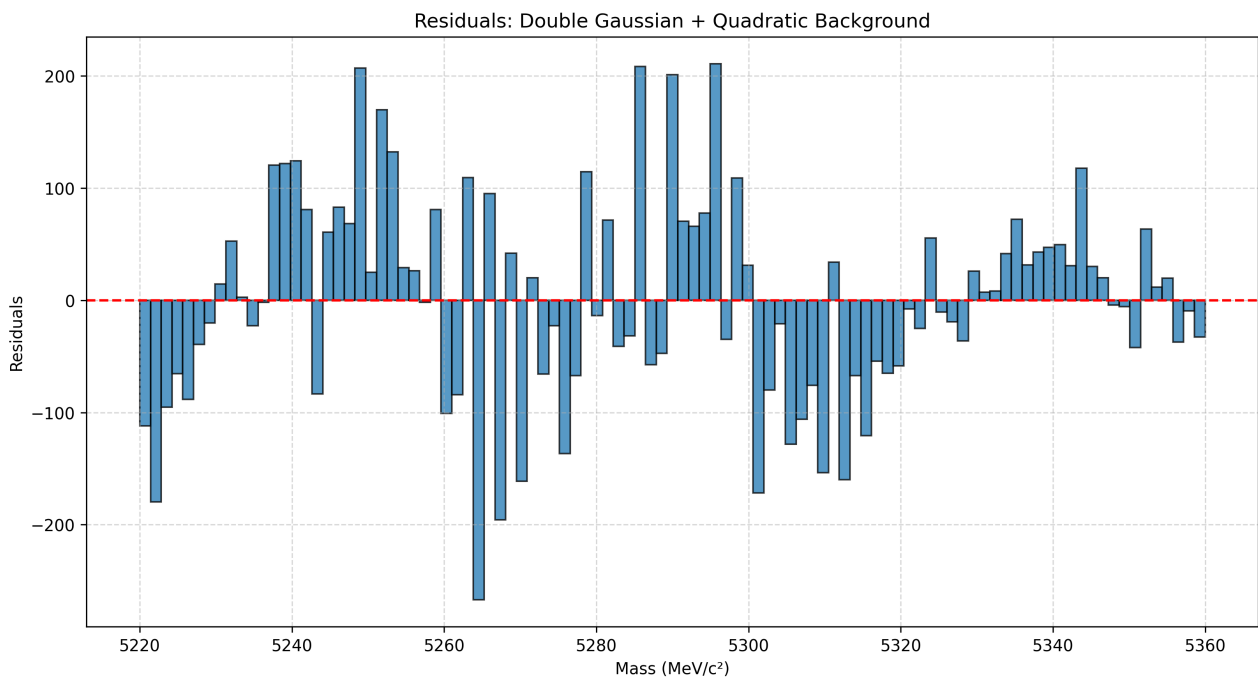


Figure 11: Raw residuals for the double-Gaussian plus quadratic-background model.

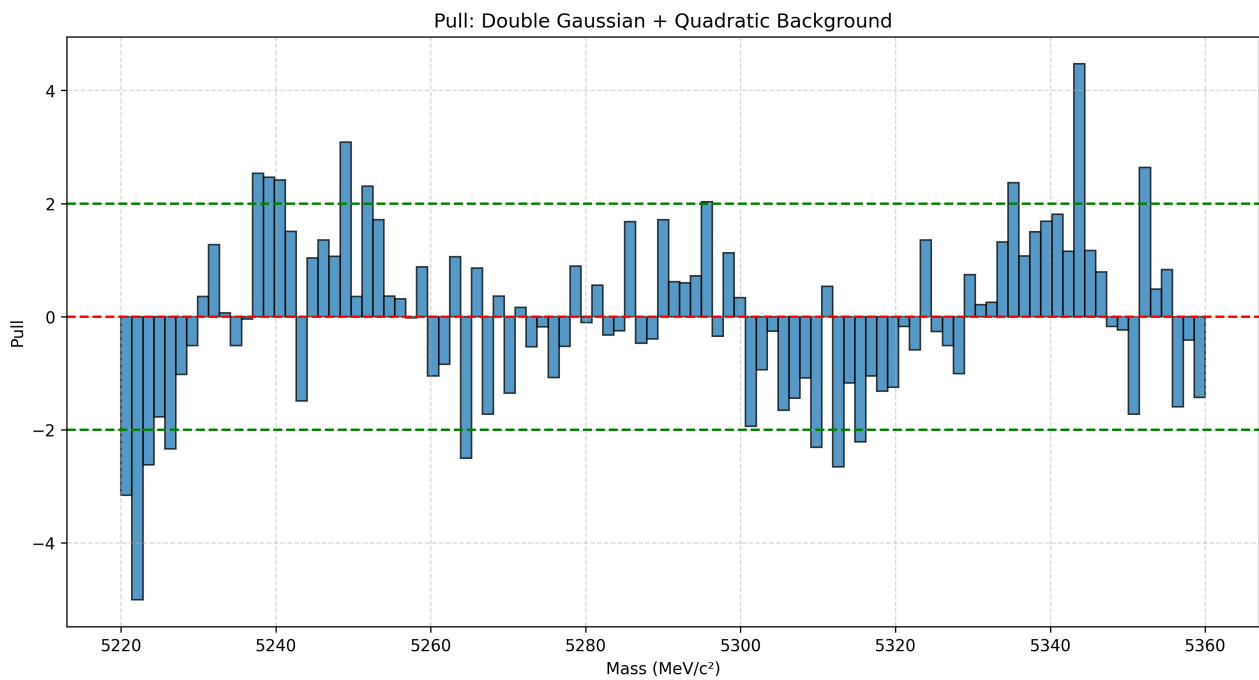


Figure 12: Pearson-style pulls for the double-Gaussian plus quadratic-background model.

For this model,

$$\chi^2 = 225.11.$$

There were eight fitted parameters, giving

$$\nu = 99 - 8 = 91.$$

The resulting diagnostic was

$$\chi_{\text{red}}^2 = 2.47.$$

This was substantially lower than either of the single-Gaussian models.

Comparing the three Analysis 4 models

Table 9: Goodness-of-fit diagnostics for the three models tested in Analysis 4.

Model	Parameters	χ^2	χ^2/ν
Single Gaussian + linear background	5	1102.49	11.73
Single Gaussian + quadratic background	6	568.97	6.12
Double Gaussian + quadratic background	8	225.11	2.47

The sequence showed a clear improvement:

$$11.73 \longrightarrow 6.12 \longrightarrow 2.47.$$

The first improvement occurred when the background was allowed to curve.

The second, larger improvement occurred when the visible peak was no longer forced to follow a single Gaussian width.

The fitted peak centre remained remarkably similar across all three models:

Table 10: Fitted peak centre across the Analysis 4 models.

Model	Fitted centre (MeV/ c^2)
Gaussian + linear	5279.7373
Gaussian + quadratic	5279.6380
Double Gaussian + quadratic	5279.6546

The centres differ by only about $0.1 \text{ MeV}/c^2$ across these three exploratory models.

I do not interpret this as a precision measurement of the $B^\pm$ mass, but it was useful to observe that the fitted location of the visible peak was relatively stable even while the model shape changed.

Why I stopped here

The final reduced diagnostic,

$$\chi^2/\nu = 2.47,$$

is still larger than one.

Therefore, even the double-Gaussian plus quadratic-background model does not perfectly describe the histogram within the simple statistical picture used here.

However, I deliberately stopped rather than continue adding cubic backgrounds, more Gaussian components or additional free parameters until the residuals became small.

A more complicated model is not automatically a better analysis

Adding enough free parameters can make a function follow almost any finite dataset more closely. The purpose of this project was to learn how model assumptions affect a fit, not to minimise the residuals at any cost.

The final model is therefore treated as the best of the small set of models I tested, not as the correct or final physical description of the LHCb mass distribution.

What Analysis 4 does and does not establish

Supported by this exercise	Not established by this exercise
A linear background describes the selected histogram poorly according to the residual and Pearson-style goodness-of-fit diagnostic.	The physical background in the LHCb sample is known to follow a specific polynomial function.
Adding a quadratic term substantially reduced the diagnostic χ^2/ν .	The quadratic model is statistically or physically proven to be the correct background model.
Replacing the single Gaussian by two Gaussians sharing one centre reduced χ^2/ν further, from 6.12 to 2.47.	The two Gaussian components correspond to two separate physical processes.
The fitted centre remained close to 5279.6–5279.7 MeV/ c^2 across the tested models.	The quoted covariance uncertainties constitute a full experimental uncertainty on the $B^\pm$ mass.
Residuals and pulls reveal model deficiencies that are difficult to see in the main histogram alone.	The final model provides a complete precision description of the data.

What I learned

1. Branches from the same ROOT tree must remain aligned. Removing entries independently from related arrays can cause quantities from different candidates to be compared incorrectly.
2. A shared Boolean mask makes it possible to apply several validity conditions while preserving the relationship between variables.
3. A cut is computationally a condition that decides which rows of the candidate sample remain.
4. Overall candidate retention alone is not enough to judge whether a selection is useful.
5. Comparing retention separately in each mass bin showed information that was difficult to see from overlaid histograms.
6. A stricter selection is not automatically a better selection. It can reject more candidates away from the peak while also losing more candidates near the peak.
7. A histogram can be modelled as the sum of several mathematical components rather than with a single function.
8. A constant, linear and quadratic background make different assumptions about the shape of the candidate distribution underneath and around the peak.
9. Residuals are useful because a fit that appears visually good can still contain systematic structure.
10. Pulls scale residuals by an estimate of the expected counting fluctuation, making discrepancies between bins easier to compare.
11. A goodness-of-fit diagnostic provides a quantitative way to compare models rather than selecting one only because it looks better.
12. Improving the background model did not solve all of the fit structure. Allowing the peak itself to have two Gaussian widths produced a much larger additional improvement.
13. Adding parameters indefinitely would risk fitting fluctuations rather than learning anything useful about the structure of the dataset.

- 14.** The covariance-derived parameter uncertainties belong to the chosen mathematical fit. They do not automatically include model, detector, calibration or systematic uncertainty.

AI use declaration

I used ChatGPT extensively as a tutor and guide throughout this continuation.

After reproducing the basic CERN and REANA mass-distribution example in the first part of the project, I did not know what a sensible next analysis would be.

I therefore asked ChatGPT to suggest tasks that would allow me to continue working with the same ROOT file while learning additional data-analysis techniques.

For Analysis 3, ChatGPT suggested investigating the ratio of

`Bplus_ENDVERTEX_CHI2`

to

`Bplus_ENDVERTEX_NDOF`.

I had not previously encountered these branches and did not independently know how they should be used in an LHCb analysis.

ChatGPT helped me understand:

- why related ROOT branches must be filtered using the same mask,
- why independently removing invalid values could break candidate alignment,
- what a candidate-selection cut means computationally,
- why several thresholds should be compared,
- why overall retention was not sufficient,
- and why retention should also be calculated as a function of reconstructed mass.

For Analysis 4, ChatGPT suggested beginning with a Gaussian peak and linear background and then using the residuals to decide whether the model needed to be modified.

Through that process, ChatGPT helped me understand:

- how a signal-plus-background model can be constructed as the sum of several functions,
- why centring the background polynomial around a fixed reference value such as 5280 can make the parameterisation easier to work with,
- how a linear and quadratic background differ,
- how raw residuals are calculated,
- how a Pearson-style pull can be constructed,

- how a simple chi-squared-per-degree-of-freedom diagnostic can be calculated,
- why the diagnostic must be interpreted cautiously when the fit itself was performed using unweighted `curve_fit`,
- why continuing to increase the polynomial order purely to reduce residuals could lead to overfitting,
- and why testing a double-Gaussian peak was a reasonable next exploratory model once the quadratic background still left systematic residual structure.

The model sequence used in this report was therefore strongly guided by ChatGPT. I did not independently know in advance that these were the appropriate models to test.

I wrote and ran the Python analysis, inspected the numerical outputs, constructed the plots, implemented the cuts and model functions, calculated the retention ratios, residuals, pulls and goodness-of-fit diagnostics, and checked how the results changed as I modified the code.

When I encountered coding errors, I used ChatGPT to understand what the error meant and then corrected the Python implementation.

ChatGPT also helped me organise and edit this report.

This project should therefore be understood as a guided learning exercise in coding and data analysis rather than as an independent scientific discovery or a validated LHCb measurement.

The final responsibility for the code, figures, reported numerical outputs and limitations is mine.

Data, code and sources

Tools used:

Python 3.11, uproot, NumPy, Matplotlib and SciPy.

Dataset:

- $B^\pm \rightarrow J/\psi(\rightarrow \mu^+\mu^-)K^\pm$ Ntuples 4530 — 2017 Magnet Up
- Parent collection DOI: [10.7483/OOPENDATA.LHCB.EBYF.C7OY](https://doi.org/10.7483/OOPENDATA.LHCB.EBYF.C7OY)

Original analysis example:

- REANA example: LHCb Run 2 $B^\pm \rightarrow J/\psi K^\pm$ analysis

Useful documentation:

- [uproot documentation](#)
- NumPy documentation: `numpy.isfinite`
- NumPy documentation: `numpy.histogram`
- Matplotlib documentation: `matplotlib.pyplot.hist`
- SciPy documentation: `scipy.optimize.curve_fit`

Code:

The complete Python code will be made available on GitHub and linked through the Data Analysis page of my website.

This is a guided learning project, not a peer-reviewed particle-physics study. It documents my attempt to continue beyond a basic CERN example by applying a candidate-selection condition, testing several peak-plus-background models and using residuals and simple goodness-of-fit diagnostics to compare them.